SHES4 USER MANUAL

Energy & Power, We Deliver Better.
SEONGHWA

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1. INTRODUCTION

SH engineering System, hereafter SHES4, is the engineering software developed by ourselves, which will be effectively applicable for the following engineering works;

1) Selection SH standard(catalogue) support based on basic data (i.e. load, displacement, and installation allowance etc.)
2) Generation for Bill of Material
3) Generation for DXF drawings as a result of step 1)

This program has been developed as "Standard alone Type" to operate User Personnel computer.

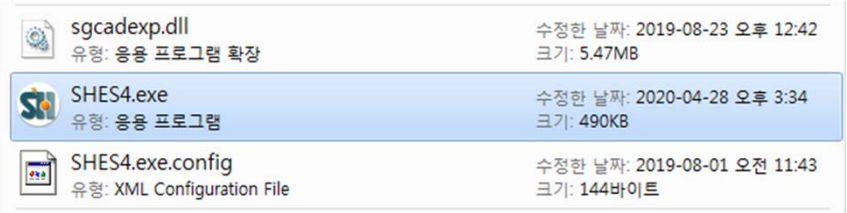
In addition, Embedded FireBird Database (.FDB) is used for database, and SHES4 will be executed by Windows program(Windows7, Windows10) under Microsoft .NET Framework 4.0 surroundings

2. Installation & Run of Program

1) Microsoft .NET Framework 4.0 should have been already installed.

2) If Microsoft .NET Framework 4.0 is not available, download and install this file from Download Center (http://www.microsoft.com/) or use installation file in NET Framework folder of CD.
(If it is already installed, it's not required installation once again.)

3) Installation and run of programs in the following steps.
   a. Download the latest program files (*.zip) from the Seonghwa Industrial website.
   b. Unzip the downloaded file to the desired location.
   c. Double-click on the "SHES4.exe" file in the decompressed folder to run it.



(Fig. 2-1)

(Fig. 2-2)

*** Caution**
If you are using an old version of SHES4, you must delete the old version and unzip the latest program before running.
If not delete the old version of SHES4, this may be the cause of the error.

3. Main Menu

3-1) Main Menu

1) Main Menu (Fig. 3-1) will be opened when running SHES4.

2) SHES4 consists of SIMPLE DESIGN and DETAIL DESIGN function.

   - SIMPLE DESIGN is used for simple selection of SeongHwa Support.

   - DETAIL DESIGN is used for generating detail drawing, it's based on the detail factors.

3) If you have any queries, please click the "CONTACT US" button from the bottom menu.

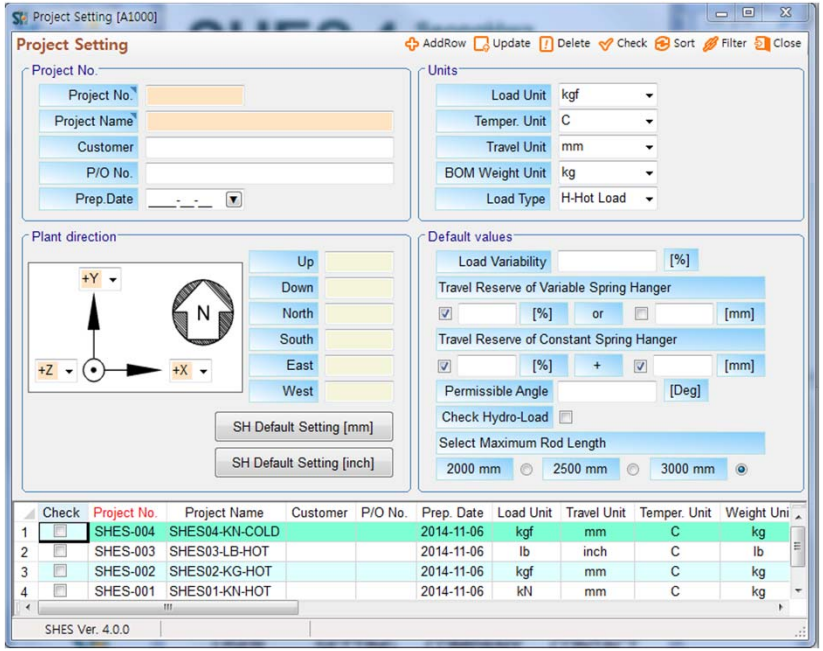
4) If you want to read SH company profile, please click the "COMPANY PROFILE" button.

(Fig. 3-1)

4. Project Setting

4-1) Project Setting

1) On Main Menu (Fig. 3-1), Click “SETTING” button and the window(Fig. 4-1) will be displayed.
2) Project no. and name are input of user choice.
3) Above 2) shall be filled. (No blank)
4) In Units field, choose values of Unit.
5) In Plant direction field, set direction of X, Y, Z.
6) If you click “SH Default Setting” button, Default Values is automatically chosen.
7) Click ‘AddRow’ button to generate project record after inputting data.
8) To modify inputted data, click the relevant record on grid, re-input, and then click “Update” button.
9) To delete record(s), click relevant Project no. on grid, and click “Delete” button. (Several Projects can be erased at once by selecting them in “Check” column.)
10) In case of checking of Hydro. Test load, click ‘Check Hydro-Load’ bottom in Default Values field



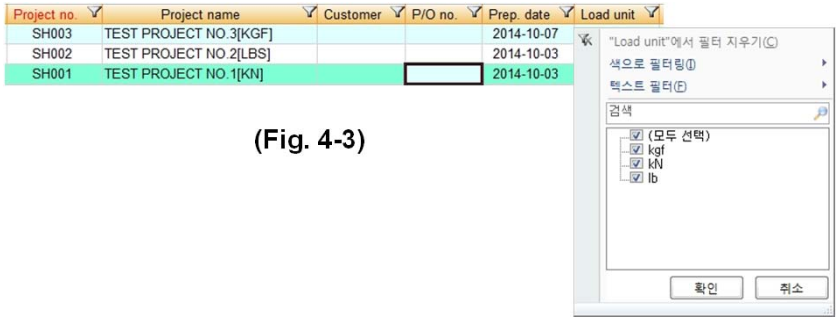
(Fig. 4-1)

11) Click the “Sort” button on the menu, and the window(Fig. 4-2) will be displayed.
Data will be sorted descending or ascending order by clicking the column.

Project no.	Project name	Customer	P/O no.	Prep. date	Load unit
SH001	TEST PROJECT NO.1[KN]			2014-10-03	kN
SH002	TEST PROJECT NO.2[LBS]			2014-10-03	lb
SH003	TEST PROJECT NO.3[KGF]			2014-10-07	kgf
SH004	TEST PROJECT NO.4(COLD)			2014-10-17	kgf

(Fig. 4-2)

12) Click the “Filter” button on the menu, and the window(Fig. 4-3) will be displayed.
Selected column shows contents of Fig. 4-3, and you can select which will be displayed.



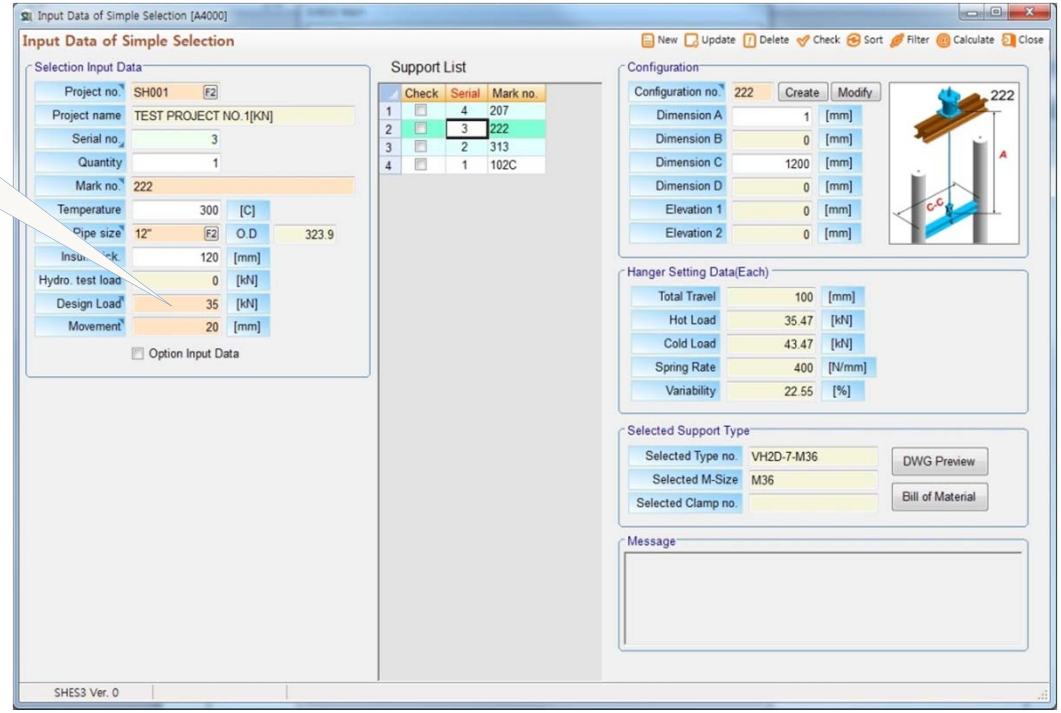
(Fig. 4-3)

5. SIMPLE DESIGN

5-1) INPUT DATA setting

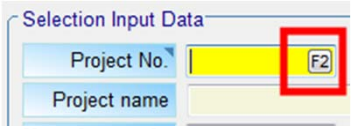
1) On Main Menu (Fig. 3-1), Click SIMPLE DESIGN button and the window (Fig. 5-1) will show-up.

"Light Pink Cell" must be filled.



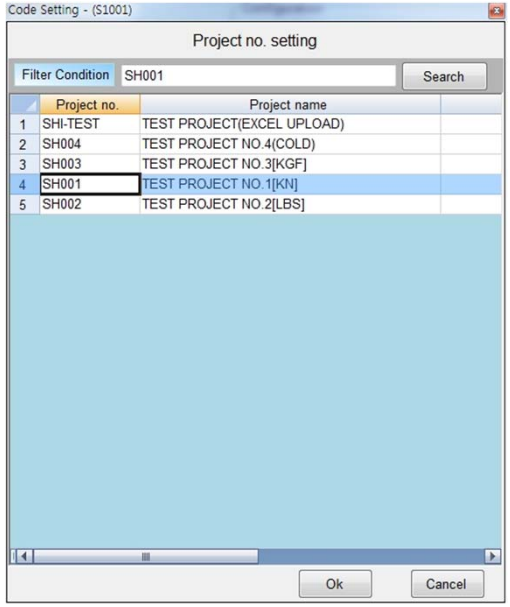
(Fig. 5-1)

2) If you click 'F2' bottom beside Project No. , Project No. setting window (Fig. 5-2) will be displayed, Select Project, among the registered Projects, and double click relevant Project No. or press enter key on grid.

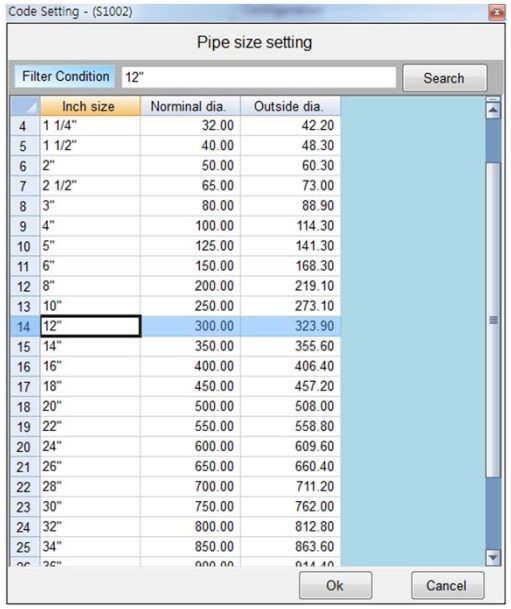


3) Mark no. should be filled. (No blank.)

4) If you click 'F2' bottom beside pipe size, Pipe size setting window (Fig. 5-3) will be displayed to set data, double click the relevant Pipe size, or press enter key on grid.
In the case of Special size, double click "S" located in bottom and input O.D. on Pipe size field. (Must be input in mm)



(Fig. 5-2)



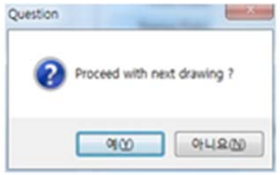
(Fig. 5-3)

5) To input Hydro. test load data, checking hydro load is needed on Project no. setting (Fig. 4-1) in advance

6) Design load and Movement should be filled. (No blank.)

7) On Configuration field, you can create / modify data by
- Inputting directly Configuration No and pressing ENTER.
  (After then, clicking Modify button for modifying support type if required)
- without inputting Configuration No, Click create button and chose Support Type

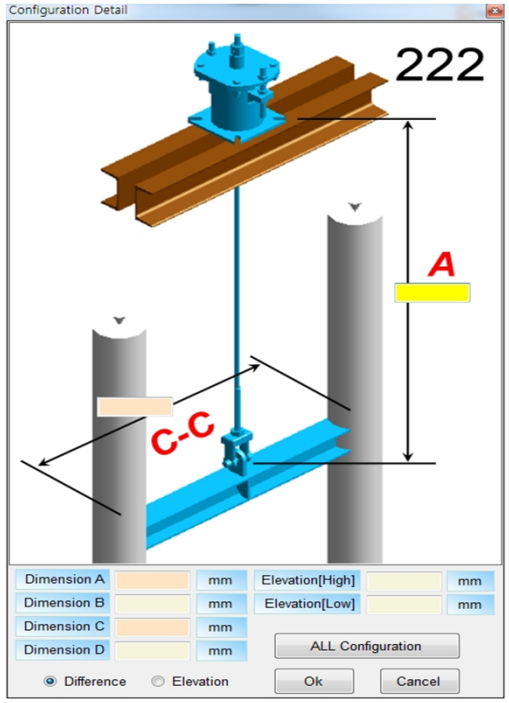
8) For example, If you input 222 Configuration No. in the field, Configuration Detail window (Fig. 5-4) will be displayed.
Input values of Dimension A and C(Red filled) and click Ok button or press enter.
The pop-up window(Fig. 5-5) will be shown to ask you for proceeding



(Fig. 5-5)

9) If you click the “No” in the pop-up window (Fig. 5-5), new record can not be generated.

10) If you select Configuration No. wrongly, press Modify button and re-select Configuration and re-input.
information on (Fig. 5-4)screen and press OK button for saving.



(Fig. 5-4)

11) If you need to modify the required data in Configuration, press Create button on the (Figure 5-1) screen to modify the data on the (Figure 5-4) screen. If you want to re-select Configuration, you can click All Configuration to re-select it

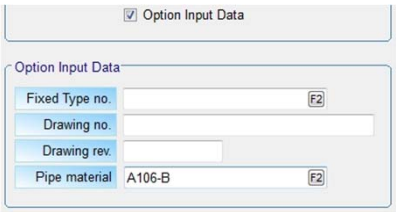
12) If you chose 'Differences' on the screen(Fig. 5-4), you can select the installation space directly. When selecting 'Elevation', you can chose installation space as your require.

13) Inputted data as Project No. on setting up stage, it will be displayed in support list.

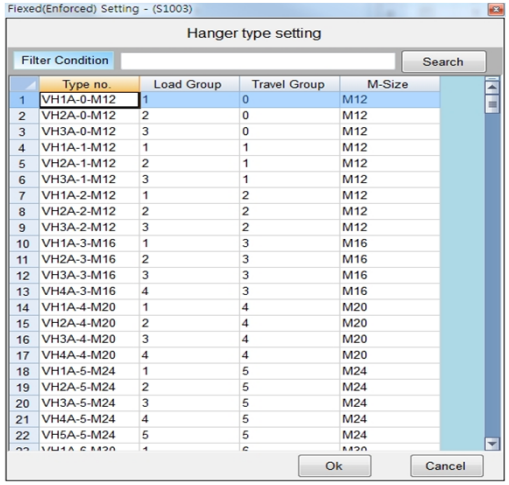
14) If you set Option Input Data (Fig. 5-1), relevant window(Fig. 5-6) will show up.

15) In the case of selecting user define Support Type, Click 'F2' in Fixed Type no. field, and choose Type no. on Hanger type setting window (Fig. 5-7)

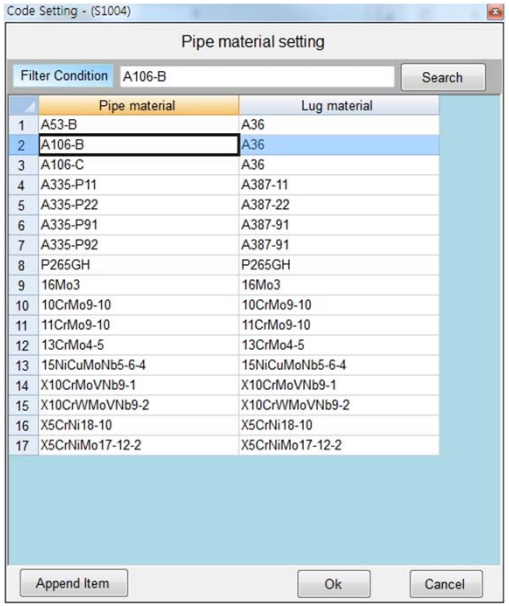
16) Same of user defined Pipe material, Click 'F2' in Pipe material field, and choose Pipe on Pipe material setting widow (Fig. 5-8)



(Fig. 5-6)



(Fig. 5-7)



(Fig. 5-8)

17) If there are no applicable material, Click “Append Item” on Pipe material setting window (Fig.5-8) and add material on Pipe Material List window(Fig. 5-9)

18) Click the “New” button, new record will appear at the bottom of the window (Fig. 5-10)

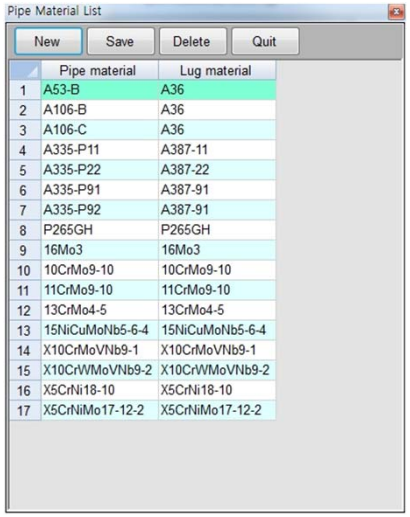
19) Input pipe material on the grid and click save button. (Fig. 5-11)

20) To delete record, select relevant record and click “Delete” button.

21) For menu Sorting or Filtering, please refer to section 11) and 12) of Program setting.

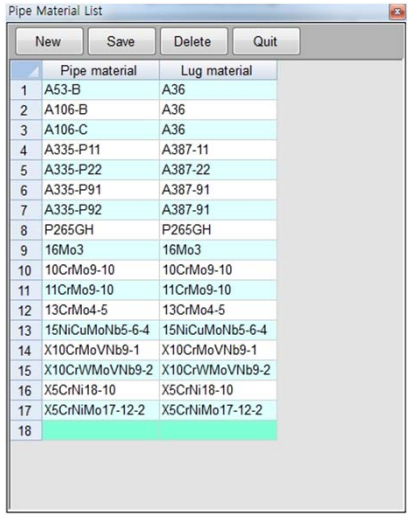
22) Click the “Calculate” button on the menu, select record on related grid will be calculated. If modify existing data which were already calculated, perform calculation after revising and updating them.

23) The calculation can be operated and chose several records.



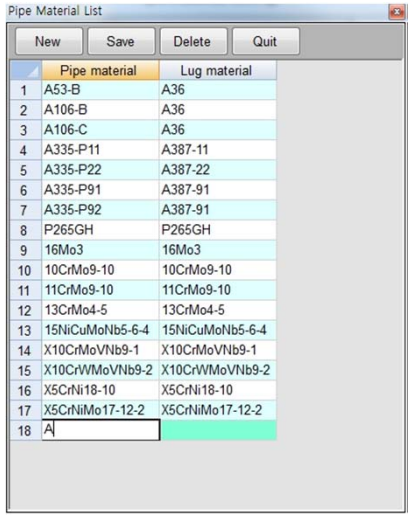
	Pipe material	Lug material
1	A53-B	A36
2	A106-B	A36
3	A106-C	A36
4	A335-P11	A387-11
5	A335-P22	A387-22
6	A335-P91	A387-91
7	A335-P92	A387-91
8	P265GH	P265GH
9	16Mo3	16Mo3
10	10CrMo9-10	10CrMo9-10
11	11CrMo9-10	11CrMo9-10
12	13CrMo4-5	13CrMo4-5
13	15NiCuMoNb5-6-4	15NiCuMoNb5-6-4
14	X10CrMoVNb9-1	X10CrMoVNb9-1
15	X10CrWMoVNb9-2	X10CrWMoVNb9-2
16	X5CrNi18-10	X5CrNi18-10
17	X5CrNiMo17-12-2	X5CrNiMo17-12-2

(Fig. 5-9)



	Pipe material	Lug material
1	A53-B	A36
2	A106-B	A36
3	A106-C	A36
4	A335-P11	A387-11
5	A335-P22	A387-22
6	A335-P91	A387-91
7	A335-P92	A387-91
8	P265GH	P265GH
9	16Mo3	16Mo3
10	10CrMo9-10	10CrMo9-10
11	11CrMo9-10	11CrMo9-10
12	13CrMo4-5	13CrMo4-5
13	15NiCuMoNb5-6-4	15NiCuMoNb5-6-4
14	X10CrMoVNb9-1	X10CrMoVNb9-1
15	X10CrWMoVNb9-2	X10CrWMoVNb9-2
16	X5CrNi18-10	X5CrNi18-10
17	X5CrNiMo17-12-2	X5CrNiMo17-12-2
18		

(Fig. 5-10)



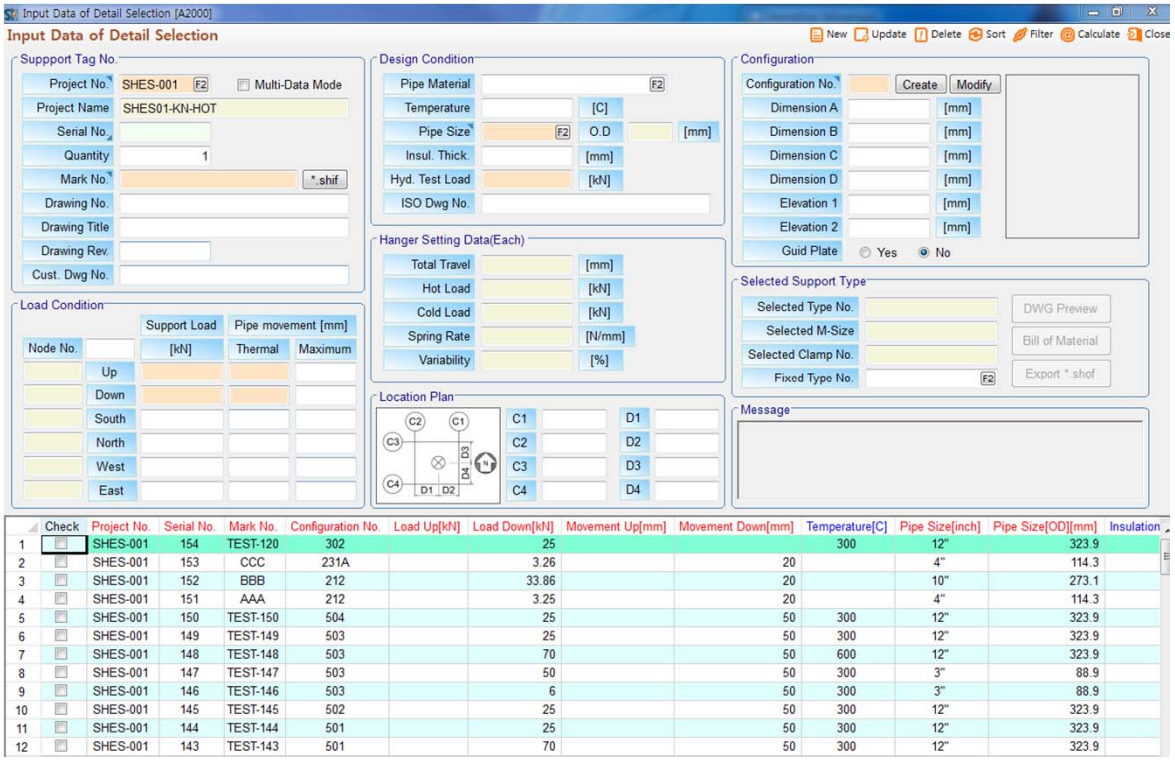
	Pipe material	Lug material
1	A53-B	A36
2	A106-B	A36
3	A106-C	A36
4	A335-P11	A387-11
5	A335-P22	A387-22
6	A335-P91	A387-91
7	A335-P92	A387-91
8	P265GH	P265GH
9	16Mo3	16Mo3
10	10CrMo9-10	10CrMo9-10
11	11CrMo9-10	11CrMo9-10
12	13CrMo4-5	13CrMo4-5
13	15NiCuMoNb5-6-4	15NiCuMoNb5-6-4
14	X10CrMoVNb9-1	X10CrMoVNb9-1
15	X10CrWMoVNb9-2	X10CrWMoVNb9-2
16	X5CrNi18-10	X5CrNi18-10
17	X5CrNiMo17-12-2	X5CrNiMo17-12-2
18	A	

(Fig. 5-11)

6. DETAIL DESIGN

6-1) INPUT DATA

1) Click DETAIL DESIGN button on the Main Menu (Fig. 3-1), a window(Fig. 6-1) will be opened.



(Fig. 6-1)

2) The selection of Project No. in the support Tag No. area refer to paragraph 2) of SIMPE DESIGN.

3) Mark no. should be filled. (No blank).

4) In Load Condition area, Support load should be properly input considering pre-set unit and direction of load. (upward/downward).

5) In Load Condition area, support thermal movement should be properly input considering pre-set unit and direction of movement(upward/downward).

6) For selecting Pipe material, pleas refer to sections 14) through 18) of SIMPLE DESIGN.

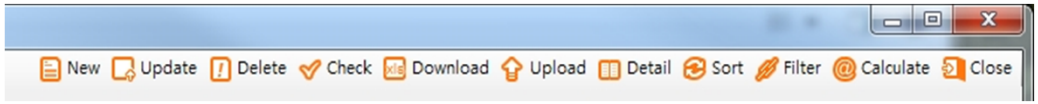
7) For selecting Pipe size, please refer to the section 5) of SIMPLE DESIGN.

8) To input Hydro. test load data, checking hydro load is needed on Project no. setting.(Fig. 4-1) in advance.

9) For setting Configuration No., please refer to sections, 8) through 11) of SIMPLE DESIGN.

10) Click “New” button on the menu to clear input data.
(If select Configuration No. without adding new record, the new record is added automatically.)

11) If you select Multi-Data Mode in (Fig. 6-1), the menus will be added and appear (Fig. 6-2).
(Check, Download, Upload, Detail menus will be added.)



(Fig. 6-2)

12) When deleting multiple items or performing a calculation, you can select or clear them collectively using Check button in the menu.

13) Error messages that occur after completion of Calculate can be found in the Message area of (Fig. 6-1), and must be checked when show up.

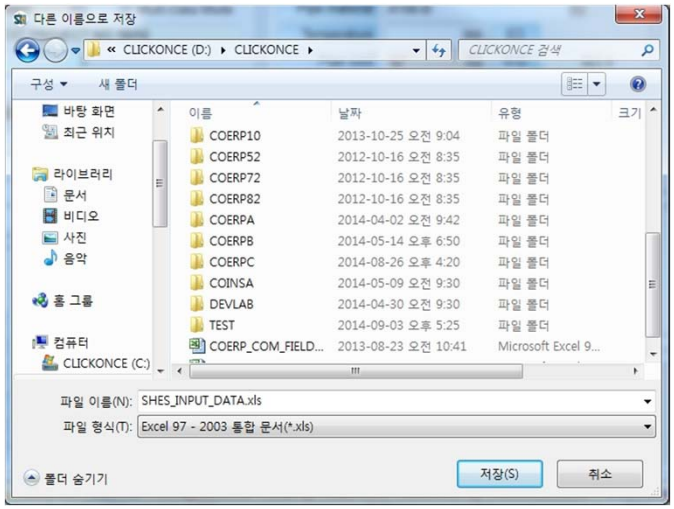
14) Click Detail button in the menu to hide the details, only grid is displayed, and click again to display details.

15) Input data can export to Excel file, click “Download” on menu and set folder and file name (Fig. 6-3).

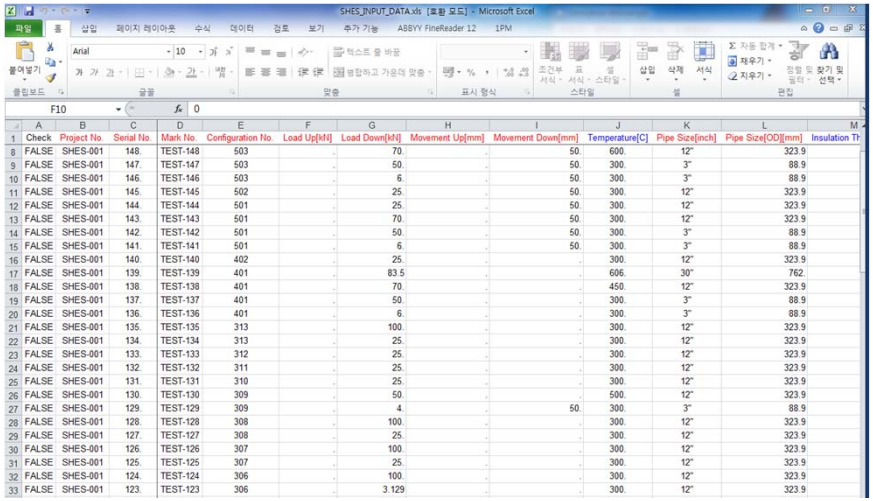
16) After downloading, you can open the Excel file in the folder (Fig. 6-4).

17) When editing values of factor in excel sheet, the form of Excel should be kept avoid any error.

18) Red columns on the Header are required data and the blue ones are optional.



(Fig. 6-3)



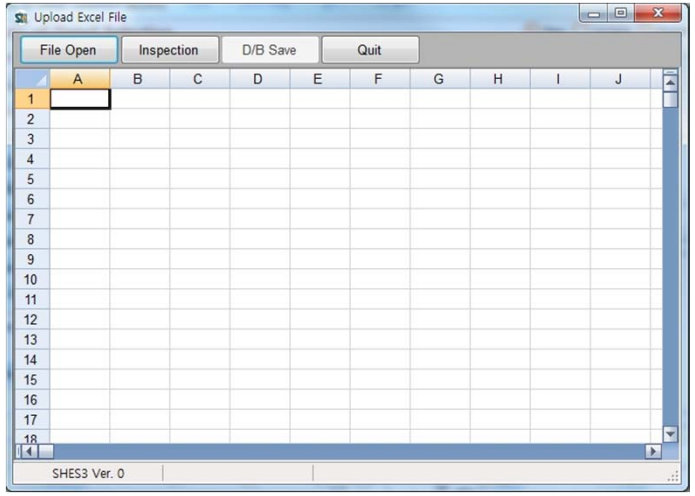
Check	Project No.	Serial No.	Mark No.	Configuration No.	Load Up[kN]	Load Down[kN]	Movement Up[mm]	Movement Down[mm]	Temperature[C]	Pipe Size[inch]	Pipe Size[OD][mm]	Insulation Th
FALSE	SHES-001	148.	TEST-148	503	.	70.	.	50.	600.	12"	323.9	
FALSE	SHES-001	147.	TEST-147	503	.	50.	.	50.	300.	3"	88.9	
FALSE	SHES-001	146.	TEST-146	503	.	6.	.	50.	300.	3"	88.9	
FALSE	SHES-001	145.	TEST-145	502	.	25.	.	50.	300.	12"	323.9	
FALSE	SHES-001	144.	TEST-144	501	.	25.	.	50.	300.	12"	323.9	
FALSE	SHES-001	143.	TEST-143	501	.	70.	.	50.	300.	12"	323.9	
FALSE	SHES-001	142.	TEST-142	501	.	50.	.	50.	300.	3"	88.9	
FALSE	SHES-001	141.	TEST-141	501	.	6.	.	50.	300.	3"	88.9	
FALSE	SHES-001	140.	TEST-140	402	.	25.	.	.	300.	12"	323.9	
FALSE	SHES-001	139.	TEST-139	401	.	83.5	.	.	606.	30"	762.	
FALSE	SHES-001	138.	TEST-138	401	.	70.	.	.	450.	12"	323.9	
FALSE	SHES-001	137.	TEST-137	401	.	50.	.	.	300.	3"	88.9	
FALSE	SHES-001	136.	TEST-136	401	.	6.	.	.	300.	3"	88.9	
FALSE	SHES-001	135.	TEST-135	313	.	100.	.	.	300.	12"	323.9	
FALSE	SHES-001	134.	TEST-134	313	.	25.	.	.	300.	12"	323.9	
FALSE	SHES-001	133.	TEST-133	312	.	25.	.	.	300.	12"	323.9	
FALSE	SHES-001	132.	TEST-132	311	.	25.	.	.	300.	12"	323.9	
FALSE	SHES-001	131.	TEST-131	310	.	25.	.	.	300.	12"	323.9	
FALSE	SHES-001	130.	TEST-130	309	.	50.	.	.	500.	12"	323.9	
FALSE	SHES-001	129.	TEST-129	309	.	4.	.	50.	300.	3"	88.9	
FALSE	SHES-001	128.	TEST-128	308	.	100.	.	.	300.	12"	323.9	
FALSE	SHES-001	127.	TEST-127	308	.	25.	.	.	300.	12"	323.9	
FALSE	SHES-001	126.	TEST-126	307	.	100.	.	.	300.	12"	323.9	
FALSE	SHES-001	125.	TEST-125	307	.	25.	.	.	300.	12"	323.9	
FALSE	SHES-001	124.	TEST-124	306	.	100.	.	.	300.	12"	323.9	
FALSE	SHES-001	123.	TEST-123	306	.	3.129	.	.	300.	12"	323.9	

(Fig. 6-4)

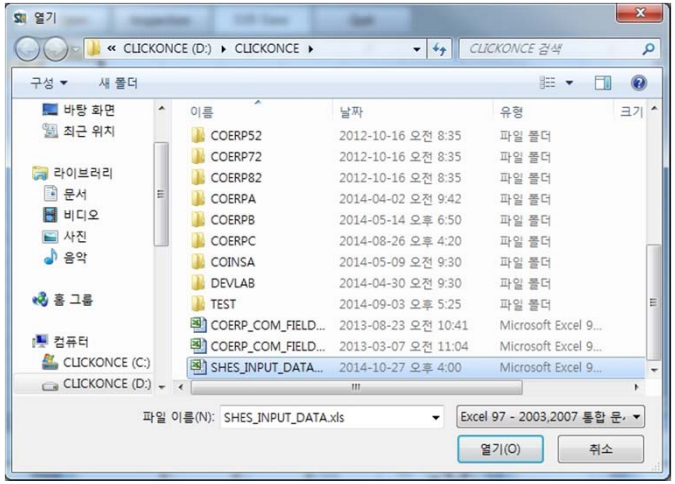
19) Click “Upload” on the menu to import excel data, upload Excel File window (Fig. 6-5) will be shown.

20) Uploading excel file must be closed before you upload.

21) Click the File Open button on the screen (Fig. 6-5) to activate the screen, and select the appropriate Excel file to open it displays (Fig. 6-7)

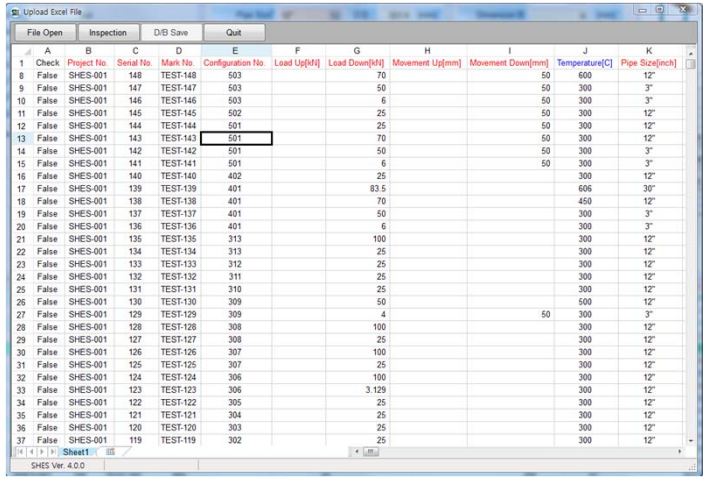


(Fig. 6-5)



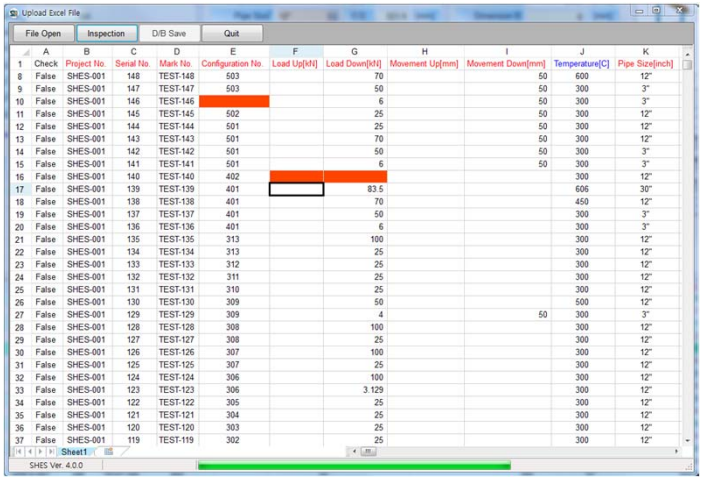
(Fig. 6-6)

22) To inspect uploaded excel file, click “Inspection” button on the window (Fig. 6-7).
23) If required data are missing, relevant cells will turn Red Orange (Fig. 6-8).
24) If there is no error, “DB Save” button will be activated (Fig. 6-9).
25) Click “DB Save” button to finalize uploading excel data.



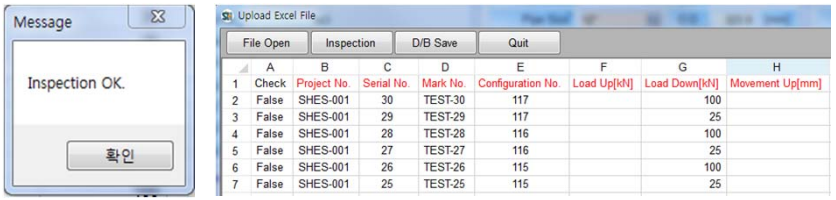
	A	B	C	D	E	F	G	H	I	J	K
1	Check	Project No.	Serial No.	Mark No.	Configuration No.	Load Up[kN]	Load Down[kN]	Movement Up[mm]	Movement Down[mm]	Temperature[C]	Pipe Size[inch]
8	False	SHES-001	148	TEST-148	503		70		50	600	12"
9	False	SHES-001	147	TEST-147	503		50		50	300	3"
10	False	SHES-001	146	TEST-146	503		6		50	300	3"
11	False	SHES-001	145	TEST-145	502		25		50	300	12"
12	False	SHES-001	144	TEST-144	501		25		50	300	12"
13	False	SHES-001	143	TEST-143	501		70		50	300	12"
14	False	SHES-001	142	TEST-142	501		50		50	300	3"
15	False	SHES-001	141	TEST-141	501		6		50	300	3"
16	False	SHES-001	140	TEST-140	402		25			300	12"
17	False	SHES-001	139	TEST-139	401		83.5			606	30"
18	False	SHES-001	138	TEST-138	401		70			450	12"
19	False	SHES-001	137	TEST-137	401		50			300	3"
20	False	SHES-001	136	TEST-136	401		6			300	3"
21	False	SHES-001	135	TEST-135	313		100			300	12"
22	False	SHES-001	134	TEST-134	313		25			300	12"
23	False	SHES-001	133	TEST-133	312		25			300	12"
24	False	SHES-001	132	TEST-132	311		25			300	12"
25	False	SHES-001	131	TEST-131	310		25			300	12"
26	False	SHES-001	130	TEST-130	309		50			500	12"
27	False	SHES-001	129	TEST-129	309		4		50	300	3"
28	False	SHES-001	128	TEST-128	308		100			300	12"
29	False	SHES-001	127	TEST-127	308		25			300	12"
30	False	SHES-001	126	TEST-126	307		100			300	12"
31	False	SHES-001	125	TEST-125	307		25			300	12"
32	False	SHES-001	124	TEST-124	306		100			300	12"
33	False	SHES-001	123	TEST-123	306		3.129			300	12"
34	False	SHES-001	122	TEST-122	305		25			300	12"
35	False	SHES-001	121	TEST-121	304		25			300	12"
36	False	SHES-001	120	TEST-120	303		25			300	12"
37	False	SHES-001	119	TEST-119	302		25			300	12"

(Fig. 6-7)



	A	B	C	D	E	F	G	H	I	J	K
1	Check	Project No.	Serial No.	Mark No.	Configuration No.	Load Up[kN]	Load Down[kN]	Movement Up[mm]	Movement Down[mm]	Temperature[C]	Pipe Size[inch]
8	False	SHES-001	148	TEST-148	503		70		50	600	12"
9	False	SHES-001	147	TEST-147	503		50		50	300	3"
10	False	SHES-001	146	TEST-146			6		50	300	3"
11	False	SHES-001	145	TEST-145	502		25		50	300	12"
12	False	SHES-001	144	TEST-144	501		25		50	300	12"
13	False	SHES-001	143	TEST-143	501		70		50	300	12"
14	False	SHES-001	142	TEST-142	501		50		50	300	3"
15	False	SHES-001	141	TEST-141	501		6		50	300	3"
16	False	SHES-001	140	TEST-140	402					300	12"
17	False	SHES-001	139	TEST-139	401		83.5			606	30"
18	False	SHES-001	138	TEST-138	401		70			450	12"
19	False	SHES-001	137	TEST-137	401		50			300	3"
20	False	SHES-001	136	TEST-136	401		6			300	3"
21	False	SHES-001	135	TEST-135	313		100			300	12"
22	False	SHES-001	134	TEST-134	313		25			300	12"
23	False	SHES-001	133	TEST-133	312		25			300	12"
24	False	SHES-001	132	TEST-132	311		25			300	12"
25	False	SHES-001	131	TEST-131	310		25			300	12"
26	False	SHES-001	130	TEST-130	309		50			500	12"
27	False	SHES-001	129	TEST-129	309		4		50	300	3"
28	False	SHES-001	128	TEST-128	308		100			300	12"
29	False	SHES-001	127	TEST-127	308		25			300	12"
30	False	SHES-001	126	TEST-126	307		100			300	12"
31	False	SHES-001	125	TEST-125	307		25			300	12"
32	False	SHES-001	124	TEST-124	306		100			300	12"
33	False	SHES-001	123	TEST-123	306		3.129			300	12"
34	False	SHES-001	122	TEST-122	305		25			300	12"
35	False	SHES-001	121	TEST-121	304		25			300	12"
36	False	SHES-001	120	TEST-120	303		25			300	12"
37	False	SHES-001	119	TEST-119	302		25			300	12"

(Fig. 6-8)



	A	B	C	D	E	F	G	H
1	Check	Project No.	Serial No.	Mark No.	Configuration No.	Load Up[kN]	Load Down[kN]	Movement Up[mm]
2	False	SHES-001	30	TEST-30	117		100	
3	False	SHES-001	29	TEST-29	117		25	
4	False	SHES-001	28	TEST-28	116		100	
5	False	SHES-001	27	TEST-27	116		25	
6	False	SHES-001	26	TEST-26	115		100	
7	False	SHES-001	25	TEST-25	115		25	

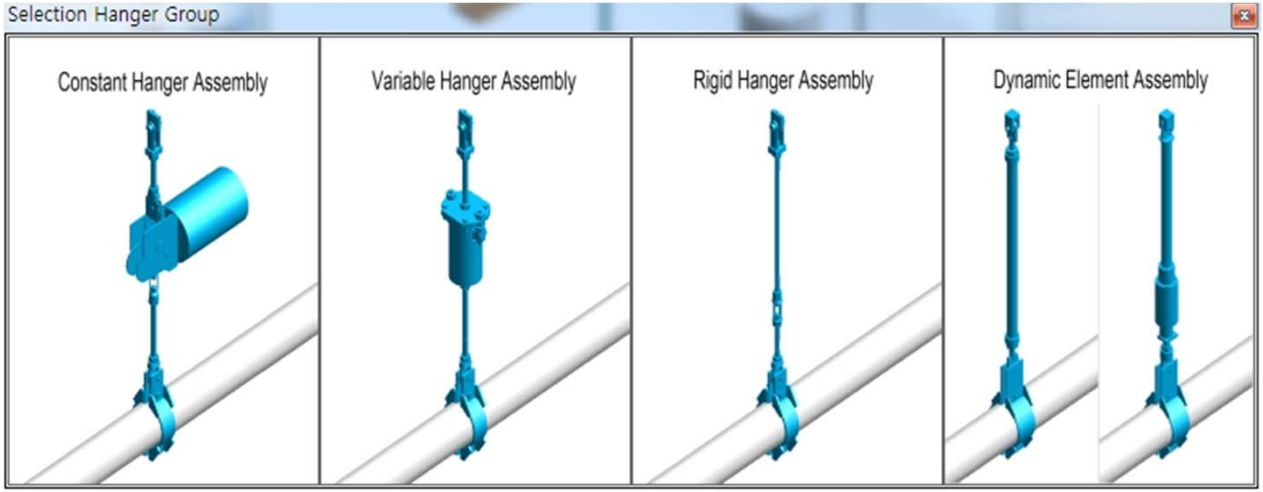
(Fig. 6-9)

7. CONFIGURATION

7-1) CONFIGURATION NO.

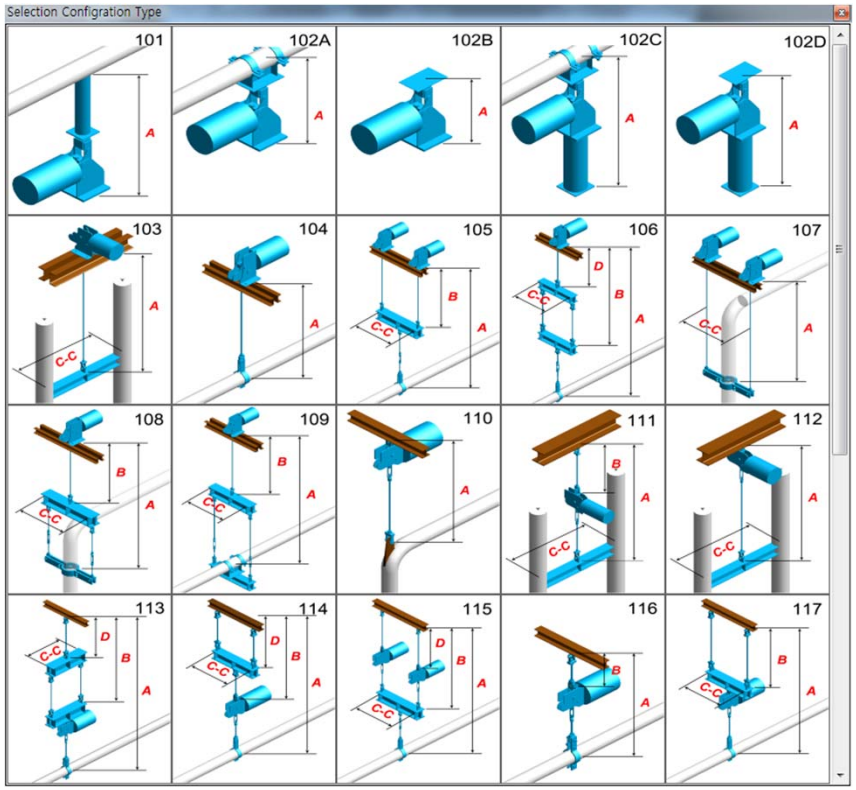
1) The selection of Configuration no. can be classified into 4 types (Fig. 7-1).
2) The number of Type no. by group is as follows.
   - Constant Hanger Assembly : 34 cases
   - Variable Hanger Assembly : 36 cases
   - Rigid Hanger Assembly : 13 cases
   - Dynamic Element Assembly : 6 cases

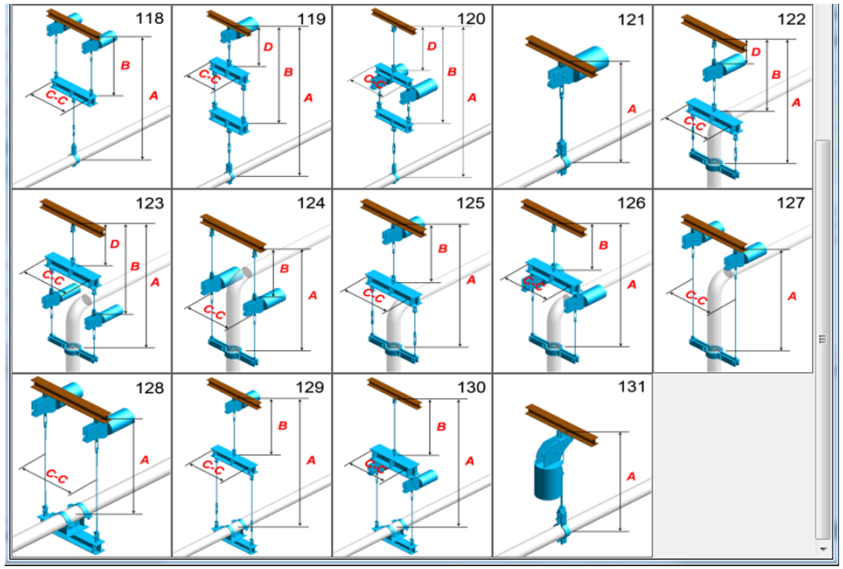
   Total number of all types: 89 cases



(Fig. 7-1)

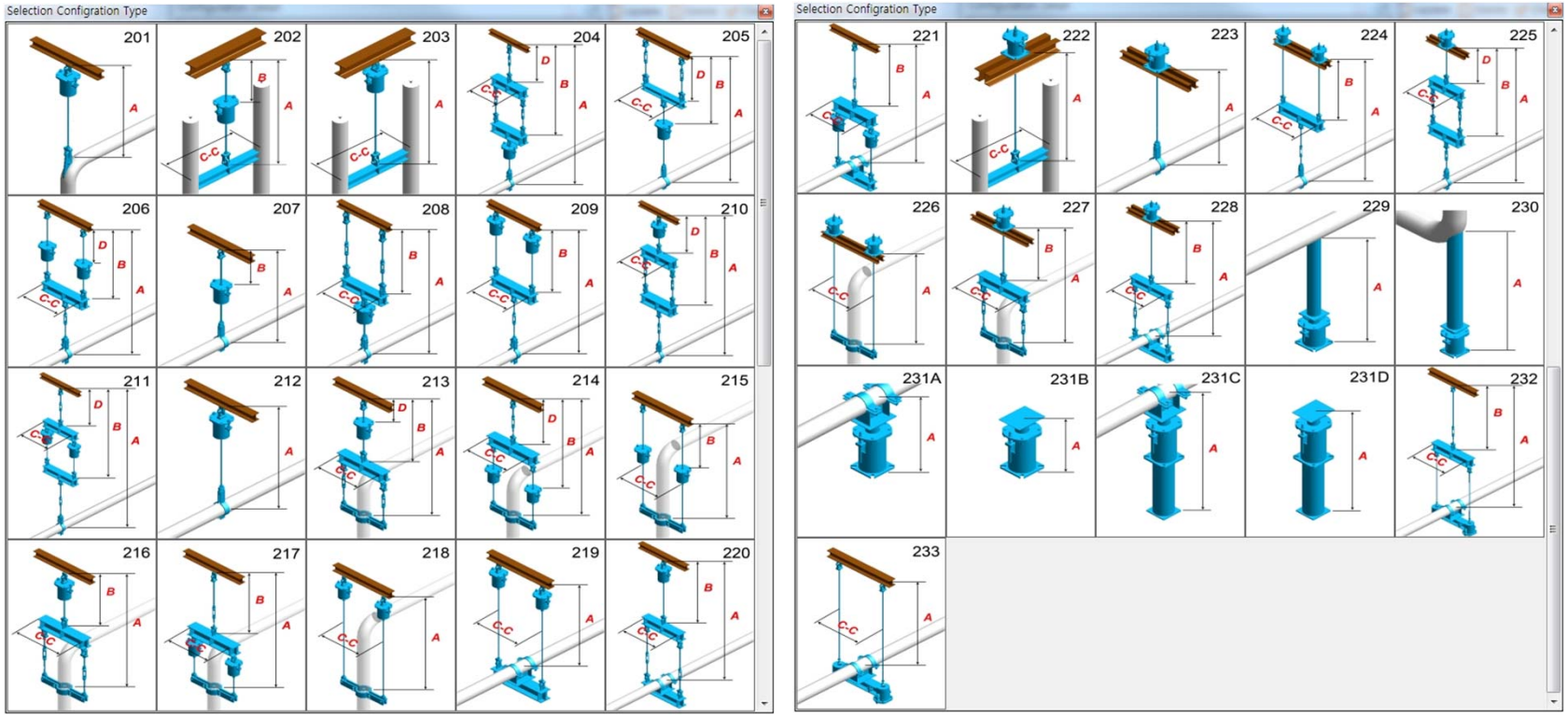
7-2) CONSTANT HANGER ASSEMBLY





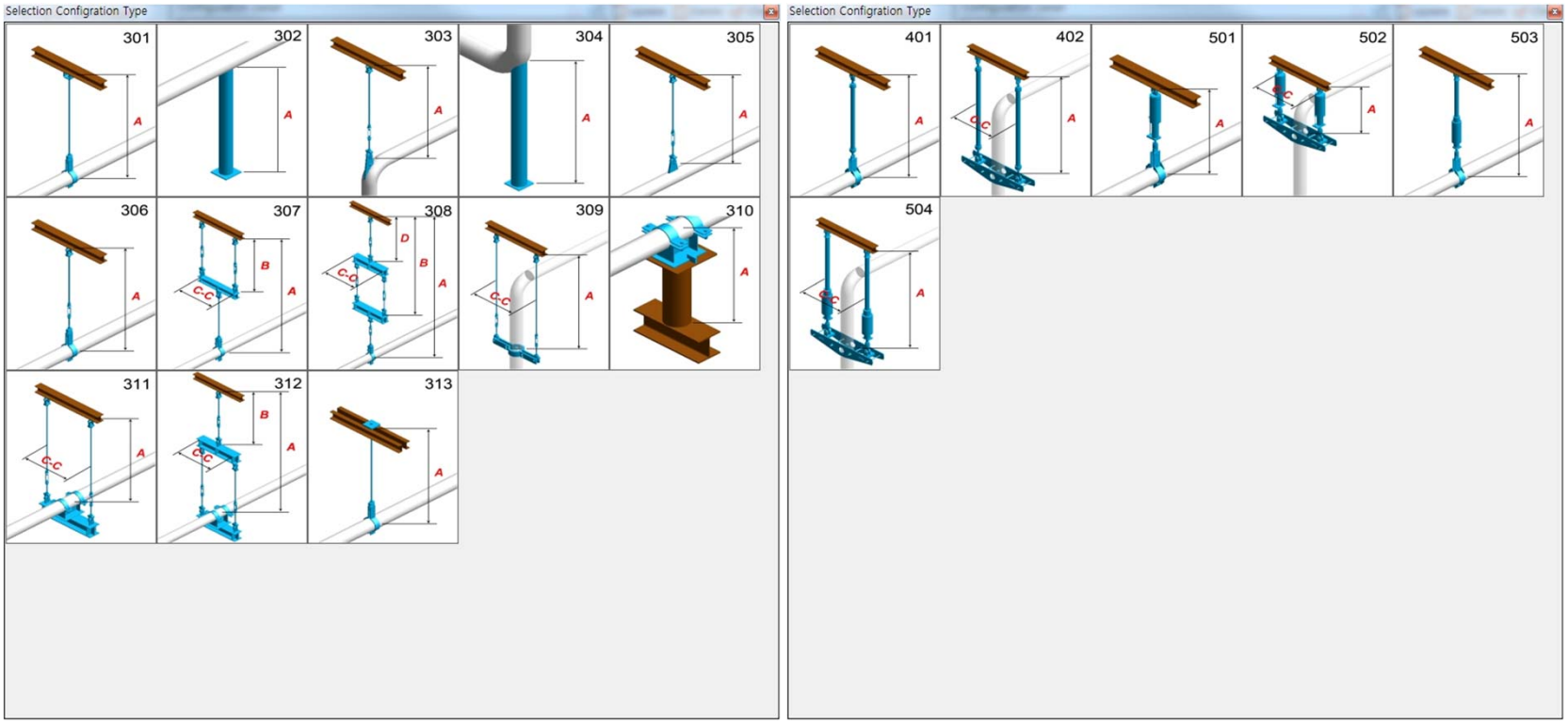
(Fig. 7-2)

7-3) VARIABLE HANGER ASSEMBLY



(Fig. 7-3)

7-4) RIGID HANGER ASSEMBLY and DYNAMIC ELEMENT ASSEMBLY

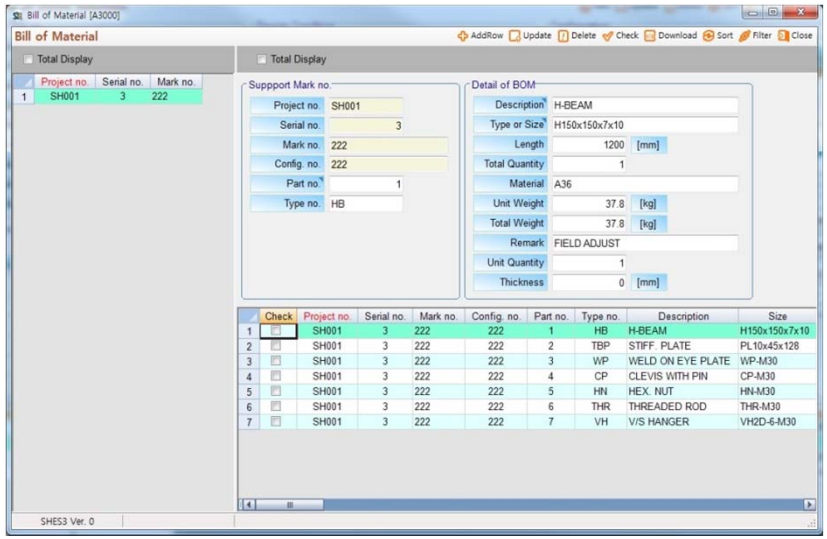


(Fig. 7-4)

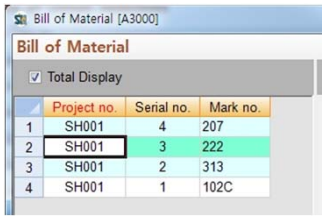
8. BILL OF MATERIAL

8-1) BOM DATA

1) Click “Bill of Material” button on INPUT DATA of SIMPLE / DETAIL DESIGN, Bill of Material window (Fig. 8-1) will be opened.
2) To view all Serial no. of the selected Project no., check “Total Display” box at the upper left
3) In order to see entire BOM of selected Project no., check “Total Display” box at top center.
4) Click “Download” on the menu to export BOM data.



(Fig. 8-1)



(Fig. 8-2)

5) If you need to add an item on BOM, you can enter the required entry items such as Part No. and Description, Type or Size, and add the item on BOM with the 'AddRow' button on the menu. (existing Part No. cannot be entered.)

6) If you want to delete an item on BOM, you can delete it by selecting the item and clicking Delete bottom from the menu.

9. DRAWING PREVIEW

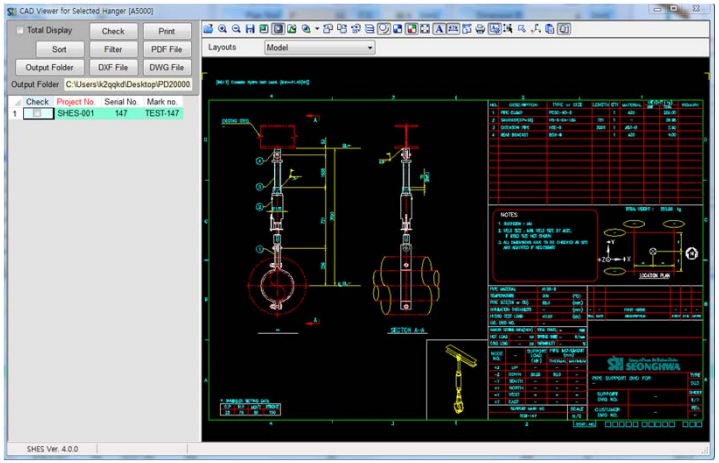
9-1) DWG PREVIEW

1) **Click “DWG Preview” button on INPUT DATA of SIMPLE / DETAIL DESIGN (Fig. 9-1)**

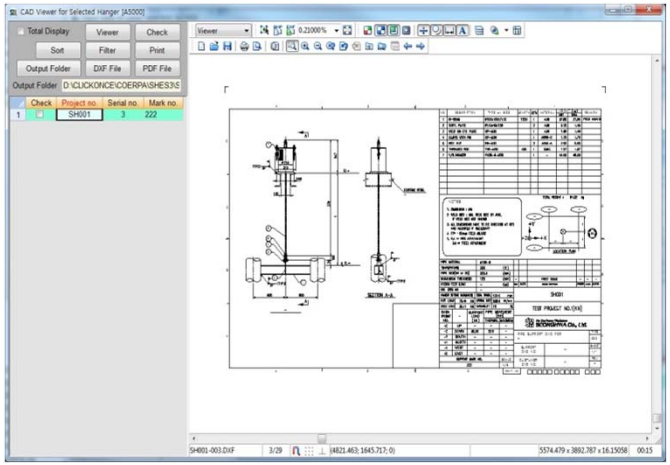
2) **Check “Total Display” box to display all Serial no. of the selected Project no.**

3) **Using icon as in (Fig. 9-2) you can change colors of contents or background of drawing.**

(Fig. 9-2)



(Fig. 9-1)

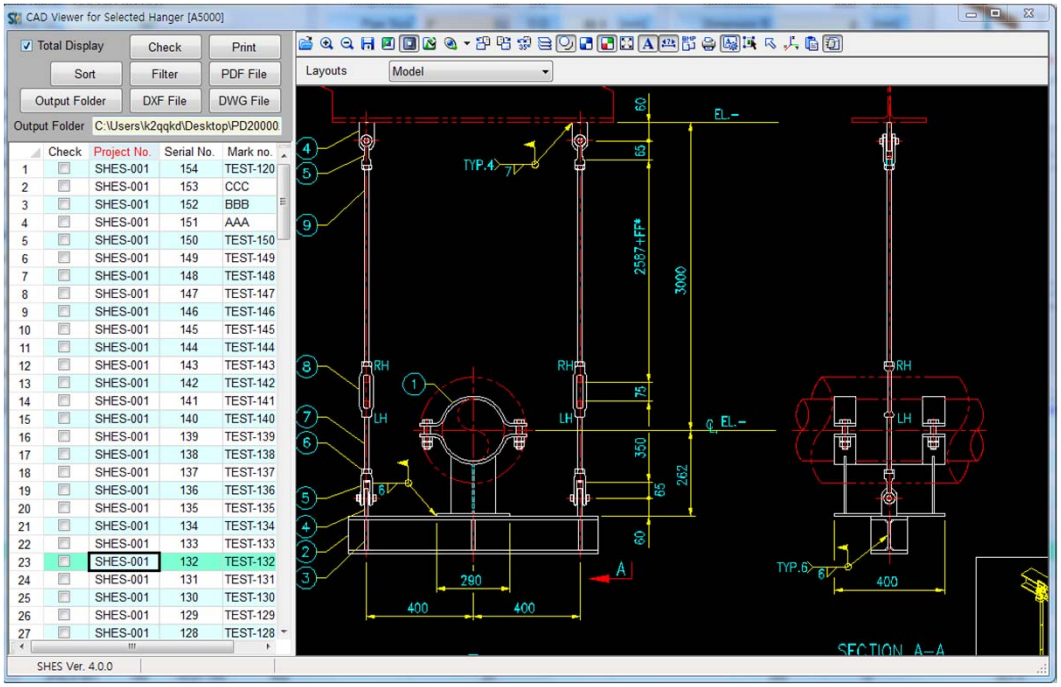


(Fig. 9-3)

4) Double click serial no. on the left of grid or click “Viewer” to check the relevant drawing.

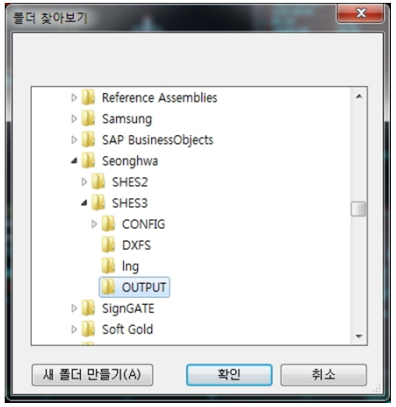
5) To zoom up drawing, drag your mouse and release. (Fig. 9-4)

6) The “Check” button can be used to print selected multi-record, or create DXF File.

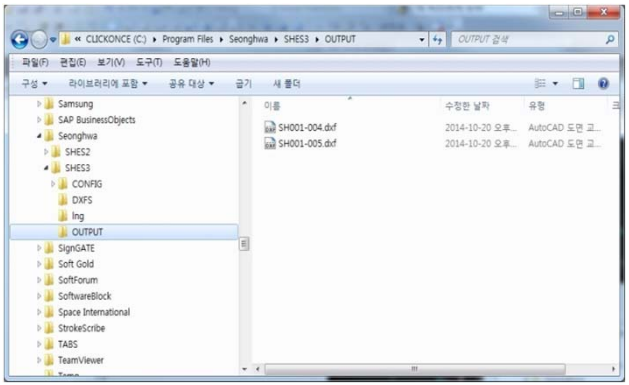


(Fig. 9-4)

7) Click “Output Folder” button to set route of saving folder. (Fig. 9-5).
8) Click “DXF File” button to create DXF File of the checked Serial no.
in the above “~OUTPUT\PROJECT NAME”.
the name of DXF file is that “Mark.no”.dxf.
9) Click “Print” button to print drawing of checked Serial no.
10) The DWG file can be created individually by selecting the desired file in the left grid.
Multi-file creation will be updated later.
11) Generating PDF file function will be updated later.



(Fig. 9-5)



(Fig. 9-6)

THANK YOU !

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